

Unit 3, Lesson 1: Proving Angle Relations



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can solve problems that use the relationships of congruent supplements, congruent complements, and vertical angles.
- ☐ I can use the relationships of congruent supplements, congruent complements, and vertical angles in proofs.



3.1.1 Warm-Up: Which One Doesn't Belong?

Four equations are shown.

$$y = 2x$$

$$y = x + 9$$

$$y = -4x + 2$$

$$y = 3x - 2$$

1. Of the four equations, choose one that does not belong.
2. Explain your reasoning for choosing the equation you did.

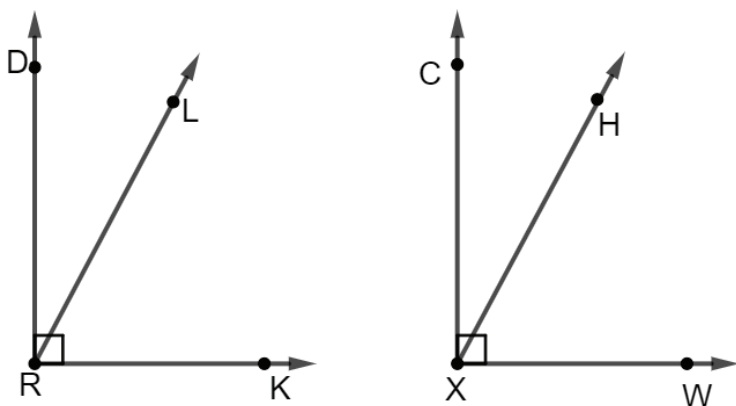


3.1.2 Exploration: Complementary Angles



Complementary angles are two angles whose measures sum to 90° .

1. Right angles DRK and CXW are shown. $\overrightarrow{RL}$ is inside $\angle DRK$ and $\overrightarrow{XH}$ is inside $\angle CXW$. $m\angle DRL = 28^\circ$ and $m\angle CXH = 28^\circ$.

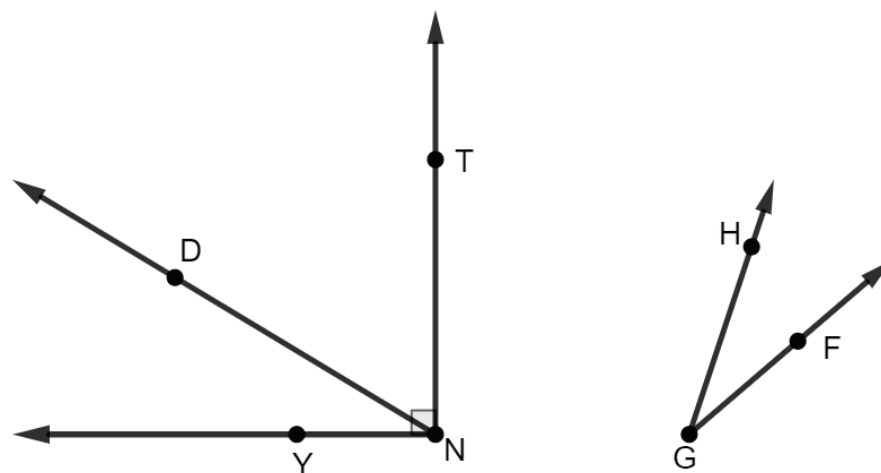


- a. Use the definition of complementary angles to explain why $\angle LRD$ and $\angle LRK$ are complementary angles.

- b. Determine the measure of $\angle LRK$.

- c. Discuss with your partner why $m\angle HXW$ must be the same as $m\angle LRK$. Summarize your discussion.

2. $\angle YNT$ and $\angle FGH$ are shown. $\overrightarrow{ND}$ is inside $\angle YNT$, such that $m\angle YND = 31^\circ$.



- a. What relationship do $\angle YND$ and $\angle DNT$ have?

b. Complete the paragraph proof.

$\angle YNT$ is a(n) ☐ acute
☐ right
☐ obtuse angle. Therefore, by

definition, $m\angle YNT = \underline{\hspace{2cm}}^\circ$. By the angle addition postulate, $m\angle YND + m\angle DNT = m\angle \underline{\hspace{2cm}}$.

Therefore, by the substitution property of equality,

$m\angle YND + m\angle DNT = \underline{\hspace{2cm}}^\circ$.

By the definition of ☐ supplementary
☐ complementary angles,

$\angle YND$ and $\angle DNT$ are ☐ supplementary
☐ complementary angles.

c. On a piece of patty paper, trace $\angle YND$. Determine which rigid transformations can be used to map $\angle YND$ onto $\angle FGH$.

d. Use patty paper or a compass to create $\angle FGW$ so that $m\angle FGW = m\angle YNT$ and $\overrightarrow{GH}$ is inside of $\angle FGW$.

e. Complete the paragraph proof.

Since $m\angle FGW = m\angle YNT$ and $m\angle YNT = \underline{\hspace{2cm}}$, then $m\angle FGW = \underline{\hspace{2cm}}$ by the transitive property of

equality. By the angle addition postulate, $m\angle FGH + m\angle WGH = m\angle \underline{\hspace{2cm}}$. Therefore, by the transitive property of equality,

$m\angle \underline{\hspace{2cm}} + m\angle \underline{\hspace{2cm}} = 90^\circ$ and are

☐ supplementary
☐ complementary angles by the definition of

☐ supplementary
☐ complementary angles.

Since $m\angle YND + m\angle DNT = 90^\circ$, then by the transitive property of equality,

$m\angle \underline{\hspace{2cm}} + m\angle \underline{\hspace{2cm}} = m\angle \underline{\hspace{2cm}} + m\angle \underline{\hspace{2cm}}$. Since $m\angle YND = m\angle FGH$, then by the substitution property $m\angle FGH + m\angle WGH = m\angle \underline{\hspace{2cm}} + m\angle DNT$. Therefore, $m\angle \underline{\hspace{2cm}} = m\angle \underline{\hspace{2cm}}$ by the subtraction property of equality.



Congruent Complements Theorem

Complements of congruent angles are congruent.

3. Use the congruent complements theorem to complete the statements.

$\angle THM$ is complementary to $\angle LGX$ and $\angle WBK$ is complementary to $\angle MCY$.

$\angle THM$ is congruent to $\angle WBK$, so $\angle$ _____ is congruent to $\angle$ _____.

A **corollary** is a true statement that is a simple deduction from a theorem or postulate. Its proof requires only a few simple statements in addition to the proof of the original theorem or postulate.



Corollary to Congruent Complements Theorem

Complements of the same angle are congruent.

4. Use the corollary to congruent complements theorem to complete the statement.

$\angle THM$ is complementary to $\angle LGX$ and $\angle WBK$ is complementary to $\angle LGX$, so $\angle$ _____ is congruent to $\angle$ _____.

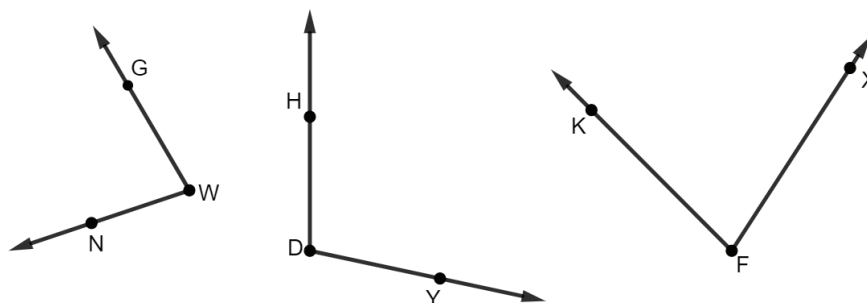


3.1.3 Exploration: Supplementary Angles



Supplementary angles are two angles whose measures sum is exactly 180° .

1. $\angle GWN$, $\angle HDY$, and $\angle KFX$ are shown. $\angle GWN$ is supplementary to $\angle HDY$. $\angle KFX$ is supplementary to $\angle HDY$.



- Determine what relationship, if any, $\angle GWN$ and $\angle KFX$ have.
- Discuss with your partner what can be concluded about $\angle GWN$ and $\angle KFX$. Summarize your discussion.

- c. Ask two classmates their conclusion about $\angle GWN$ and $\angle KFX$. Record their conclusion and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conclusion	My Summary of Their Reason	Initials

- d. Discuss your classmates' conclusions with your partner.



Congruent Supplements Theorem

Supplements of congruent angles are congruent.

2. Use the congruent supplements theorem to complete the statements.

$\angle HTM$ is supplementary to $\angle LXG$ and $\angle BKW$ is supplementary to $\angle MCY$. $\angle LXG$ is congruent to $\angle BKW$, so $\angle$ _____ is congruent to $\angle$ _____.



Corollary to Congruent Supplements Theorem

Supplements of the same angle are congruent.

3. Use the corollary to congruent supplements theorem to complete the statement.

$\angle HTM$ is supplementary to $\angle LXG$ and $\angle BKW$ is supplementary to $\angle LXG$, so $\angle$ _____ is congruent to $\angle$ _____.



3.1.4 Guided Instruction: Angles

Two adjacent angles that form a straight angle are called a linear pair.

A **linear pair** is two adjacent angles formed by two intersecting lines.



Linear Pair Postulate

If two angles form a linear pair, then they are supplementary.

1. Use the phrases in the word bank to complete the proof. Some phrases may be used more than once.

Word Bank

Definition of Linear Pair

Linear Pair Postulate

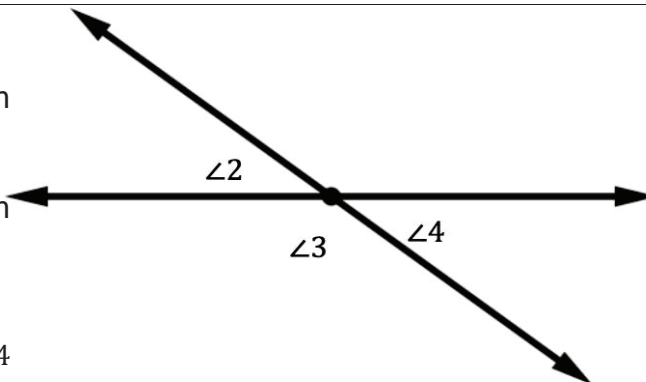
Supplements of the same angle are congruent.

Complements of the same angle are congruent.

Given:

$\angle 2$ and $\angle 3$ form a linear pair.

$\angle 3$ and $\angle 4$ form a linear pair.



Prove: $\angle 2 \cong \angle 4$

Statement	Reason
1. $\angle 2$ and $\angle 3$ form a linear pair	1. Given
2. $\angle 3$ and $\angle 4$ form a linear pair	2. Given
3. $\angle 2$ and $\angle 3$ are supplementary angles	3.
4. $\angle 3$ and $\angle 4$ are supplementary angles	4.
5. $\angle 2 \cong \angle 4$	5.



Vertical angles are the opposite angles formed when two lines intersect.

2. $\angle 2$ and $\angle 4$ are also known as _____ angles.



Vertical Angles Theorem

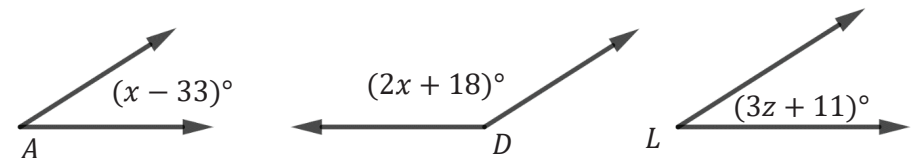
If two lines intersect, then vertical angles are congruent.



3.1.5 Try It: Solving Problems Using Angle Relationships

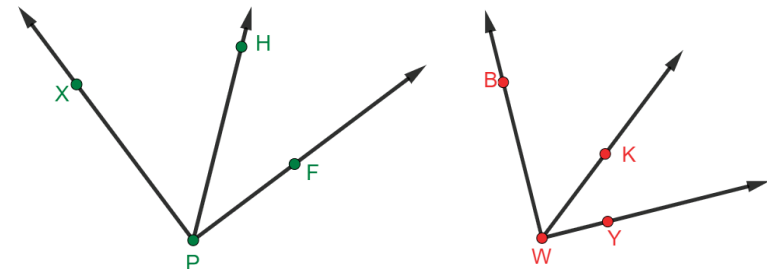
1. Three angles are shown with the given information.

$\angle A$ is supplementary to $\angle D$
 $\angle L$ is supplementary to $\angle D$



What is the value of z ?

2. Right angles $\angle XPF$ and $\angle BWY$ are shown. $\overrightarrow{PH}$ is inside $\angle XPF$, such that $m\angle HPF = 39^\circ$. $m\angle HPF = m\angle KWY$.



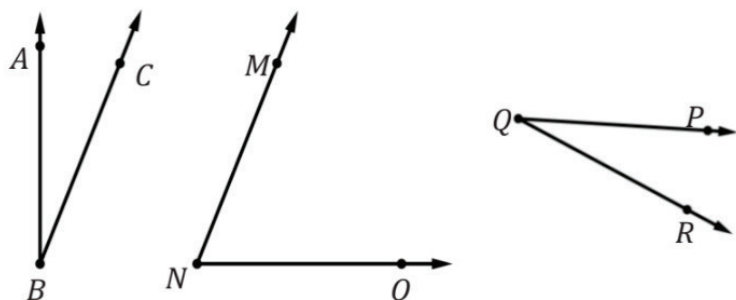
What is $m\angle BWK$?



3.1.6 Wrap-Up: Cool Down

1. Three angles are shown with the given information.

$\angle ABC$ is complementary to $\angle MNO$
 $\angle MNO$ is complementary to $\angle PQR$
 $m\angle ABC = (5x - 3)^\circ$
 $m\angle MNO = 2(x + 22)^\circ$.



- a. Find the value of x .

- b. What is $m\angle PQR$?

Unit 3, Lesson 2: Vertical and Corresponding Angles



Benchmark

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.



Learning Targets

- ☐ I can use vertical angles to solve problems.
- ☐ I can use corresponding angles formed by parallel lines and a transversal to solve problems.